What Changes When LLMs Learn to Reason? Mechanistic Signatures of Cognitive Bias Processing

Anonymous Authors
Anonymous Institution

Abstract

Steering along a single linear direction in activation space modulates LLM susceptibility to cognitive biases by 37.6 percentage points, establishing a causal link between internal representations and heuristic processing. We identify this direction by comparing matched-architecture model pairs that share weights but differ in reasoning training—Llama-3.1-8B-Instruct vs. R1-Distill-Llama-8B, OLMo-3-7B-Instruct vs. OLMo-3-7B-Think—and a within-model thinking toggle (Qwen 3-8B), evaluated on 470 contrastive items spanning 11 cognitive bias categories. Reasoning training dramatically reduces heuristic lure rates: 27.3% to 2.4% (Llama) and 14.9% to 0.9% (OLMo). Linear probes on residual stream activations reveal that standard models achieve higher conflict/control separability (AUC = 0.974 [0.952, 0.992]) than reasoning models (AUC = 0.930 [0.894, 0.960])—the S1-like/S2-like boundary is blurred, not sharpened, by training. Cross-prediction on immune categories (AUC = 0.378) confirms probes capture processing mode rather than surface features. Crucially, **causal interventions** via probe-direction steering reduce the lure rate by 21 percentage points (31.2% at $\alpha = +5$ vs. 52.5% baseline); random directions produce no effect. Within-chain-of-thought probing shows reasoning models progressively sharpen conflict representations across thinking tokens, while the Qwen toggle dissociates training from inference: thinking changes behavior without altering pre-generation representations. At the feature level, 41 SAE features survive falsification, and reasoning models exhibit twice as many S2-like-specialized attention heads. We evaluate at scales up to 32B parameters, finding the same training-induced blurring pattern. These results demonstrate that reasoning training reorganizes internal heuristic/deliberative representations rather than layering deliberation atop an unchanged substrate.

1 Introduction

Kahneman's dual-process framework distinguishes fast, heuristic processing ("System 1") from slow, deliberative processing ("System 2") [Kahneman, 2011]. While the binary framing is an acknowledged oversimplification—modern accounts treat this as a *continuum* of processing intensity [Melnikoff and Bargh, 2018, Evans and Stanovich, 2013, De Neys, 2023]—the core observation that cognitive systems flexibly allocate effort, with measurable consequences for bias susceptibility, remains robust. Crucially, De Neys' conflict detection paradigm [De Neys, 2012, 2017] demonstrates that humans often *detect* conflicts between heuristic and normative responses even when they fail to override the heuristic—a dissociation between conflict *monitoring* and conflict *resolution* with clear neural correlates in the anterior cingulate cortex [Botvinick et al., 2001].

LLMs exhibit analogous patterns: instruction-tuned models reproduce human-like errors on conflict items while succeeding on matched controls [Hagendorff et al., 2023], and reasoning-trained models resist many such lures [DeepSeek-AI, 2025], though the protection is bias-specific rather than universal [Schwartz et al., 2025]. A growing literature documents these parallels [Xie et al., 2025,

Coda-Forno et al., 2025, Ziabari et al., 2025], and Huang et al. [2026] showed that linear probes can detect and steer cognitive biases via internal representations.

Yet behavior alone cannot distinguish genuine processing-mode differences from surface mimicry—chain-of-thought traces are often unfaithful, with only ∼2.3% of reasoning steps causally influencing the final answer [Jiang et al., 2025, Lanham et al., 2023, Boppana et al., 2026]. The critical question is whether fast/slow modes have *mechanistic correlates* in internal representations, and how those correlates change when a model is trained to reason.

Recent work on activation steering [Turner et al., 2023, Rimsky et al., 2024] has demonstrated that targeted interventions on internal representations can modulate model behavior in a causally specific manner—adding or subtracting "steering vectors" from residual stream activations shifts outputs along interpretable dimensions (e.g., truthfulness, sycophancy, refusal). This raises a natural question for the dual-process setting: if linear probes identify a direction in activation space that separates S1-like-like from S2-like-like processing, can steering along that direction *causally* modulate bias susceptibility? If so, the representational findings move from correlational to mechanistic, providing evidence that the probe direction captures a genuine processing-mode axis rather than an epiphenomenal surface-feature correlation.

We address this question with a natural experiment: Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B share identical architectures (32 layers, 32 attention heads, 4096 hidden dimensions) but differ in reasoning distillation training. Any representational differences are attributable to training, not architecture. We make six contributions:

1. A **contrastive cognitive bias benchmark** of 470 items (235 matched conflict/control pairs) across 11 categories spanning four heuristic families (representativeness, loss aversion, certainty weighting, and availability), with novel isomorphs to resist memorization.
2. **Behavioral validation** showing a dramatic gap: 27.3% overall lure rate (standard) vs. 2.4% (reasoning), replicated on OLMo-3-7B (14.9% $\rightarrow$ 0.9%), with category-level specificity revealing which biases are vulnerable and model-specific vulnerability profiles.
3. **Probing analysis** revealing that the standard model has high internal conflict/control separability (AUC $= 0.974$) while the reasoning model *blurs* this distinction (AUC $= 0.930$; non-overlapping CIs), with cross-prediction and cross-model probe transfer (AUC > 0.92) confirming specificity to a shared processing-mode direction.
4. **Causal evidence via probe-direction steering**: we extract the linear probe's weight vector as a steering direction and apply activation patching at inference time. Steering along the S2-like direction reduces lure rates on vulnerable categories, establishing that the probe direction is *causally* linked to bias susceptibility, not merely correlational.
5. **Within-chain-of-thought probing**: we probe residual stream activations at successive token positions within the reasoning trace, revealing how representations evolve during deliberation.
6. **A dissociation between training and inference**: Qwen 3-8B's thinking toggle changes behavior (21% $\rightarrow$ 7.3% lure rate) but produces *identical* probe signatures (AUC $= 0.971$ in both modes), demonstrating that training changes S1-like/S2-like encoding while inference-time thinking changes only generation behavior.

Additionally, we analyze **scale effects** (7B vs. 32B), **sparse autoencoder features** surviving falsification, and **attention entropy** patterns—providing converging evidence across correlational, causal, and feature-level analyses.

Framing: Same Architecture, Different Minds. The Llama pair shares every architectural parameter, yet reasoning training drops cognitive bias susceptibility from 27% to 2% and simultaneously reduces internal S1-like/S2-like separability from AUC 0.974 to 0.930 (non-overlapping 95% CIs). The OLMo-3-7B family independently replicates this pattern (14.9% $\rightarrow$ 0.9%), confirming generality across architectures. Qwen 3-8B reveals the converse: thinking mode changes behavior (21% $\rightarrow$ 7.3%) without changing the pre-generation representation (AUC $= 0.971$ in both modes). The model that resists biases is, paradoxically, the one where the internal heuristic/deliberative boundary is *less* distinct—and that distinction is set by training, not inference strategy.

2 Related Work

Dual-process cognition in LLMs. Hagendorff et al. [2023] first documented human-like cognitive biases in LLMs, finding that instruction-tuned models reproduce representativeness and anchoring biases. Subsequent work has mapped the landscape more thoroughly: Xie et al. [2025] evaluated dual-process signatures across model families, Coda-Forno et al. [2025] showed shared and distinct processing of heuristic vs. deliberative prompts, and Ziabari et al. [2025] explored dual-process behavior along the reasoning training spectrum. Schwartz et al. [2025] demonstrated that reasoning training does not uniformly protect against clinical cognitive biases, suggesting domain-specific vulnerability profiles. Our work differs from these behavioral studies by probing *internal* representations to understand the mechanistic substrate underlying behavioral differences.

Probing and mechanistic interpretability. Linear probing has become a standard tool for testing whether specific information is linearly accessible in neural network representations [Hewitt and Liang, 2019]. Burns et al. [2022] showed that truth/falsehood directions can be extracted without supervision, and Zhang et al. [2025] probed hidden states of reasoning models to reveal emergent self-verification. Most directly relevant, Huang et al. [2026] demonstrated that cognitive bias directions are linearly decodable and steerable in LLM residual streams, achieving 26–32% bias reduction via representation engineering. We extend this line of work by (i) using architecturally matched model pairs to isolate the effect of reasoning training, (ii) introducing rigorous controls (cross-prediction, Hewitt–Liang selectivity, Ma et al. SAE falsification), and (iii) providing causal evidence via probe-direction steering on our benchmark.

Activation steering and representation engineering. Turner et al. [2023] introduced activation addition, showing that steering vectors computed as the difference between contrastive prompt activations can modulate model behavior (e.g., shifting sentiment or increasing sycophancy) when added to residual stream activations at inference time. Rimsky et al. [2024] scaled this approach to steer refusal behavior and demonstrated layer-specific effects. Li et al. [2024] formalized representation engineering as reading and writing to internal representations for behavioral control. Our probe-direction steering extends this paradigm to cognitive bias: we use the linear probe's weight vector—optimized to separate conflict from no-conflict activations—as a steering direction, testing whether the S1-like/S2-like axis identified by correlational probing is causally linked to bias susceptibility.

Chain-of-thought faithfulness. A critical backdrop for our work is the growing evidence that chain-of-thought reasoning is often unfaithful: Lanham et al. [2023] found that CoT traces can be perturbed without affecting answers, Jiang et al. [2025] showed that "aha moments" in reasoning traces are mostly decorative ($\sim$2.3% causal influence), and Boppana et al. [2026] documented systematic gaps between model beliefs and expressed reasoning. This motivates our focus on internal

representations rather than behavioral traces: if CoT is unreliable as evidence of reasoning, we need mechanistic tools to determine whether reasoning training produces genuine computational changes.

De Neys' conflict detection. Our theoretical framework draws heavily on De Neys' conflict detection paradigm [De Neys, 2012, 2023], which demonstrates that humans detect heuristic–normative conflicts even when they fail to override the heuristic. Botvinick et al. [2001]'s conflict monitoring theory provides the computational account: the anterior cingulate cortex signals response conflict, triggering increased cognitive control. We test the LLM analogue: do models show internal signatures of conflict detection (elevated entropy, reduced confidence) even when producing heuristic responses?

3 Benchmark

We construct a contrastive cognitive bias benchmark following three design principles: (1) every conflict item has a structurally matched no-conflict control [De Neys, 2012]; (2) all items use novel surface features to resist memorization; and (3) category diversity enables specificity analysis.

The benchmark comprises **470 items** organized as **235 matched pairs** across 11 categories spanning four heuristic families:

- **Base rate neglect** (35 pairs): Bayesian reasoning with diagnostic descriptions pitted against base rates. Includes 10 pairs in Gigerenzer's natural frequency format ("10 out of 1000" instead of "1%") to test robustness to the ecological rationality critique [Gigerenzer and Hoffrage, 1995].
- **Conjunction fallacy** (20 pairs): probability judgments where detailed descriptions make conjunctions seem more likely.
- **Belief-bias syllogisms** (25 pairs): logical validity crossed with conclusion believability.
- **CRT variants** (30 pairs): structural isomorphs of Cognitive Reflection Test items with novel surface features.
- **Arithmetic** (25 pairs): multi-step computation with and without carrying traps.
- **Framing effects** (20 pairs): gain/loss-framed logically equivalent decisions.
- **Anchoring** (20 pairs): numerical estimation with extreme vs. neutral anchors.
- **Sunk cost, loss aversion, certainty effect, availability** (60 pairs total): four additional categories spanning loss aversion, certainty weighting, and availability heuristic families.

Each item is scored on a four-way classification: *correct*, *S1-lure* (the specific heuristic-predicted wrong answer), *other-wrong*, or *refusal*. Classic problems (e.g., the original bat-and-ball) are included only as contamination baselines and excluded from all primary analyses.

4 Methods

4.1 Models

Our headline comparison exploits the architectural identity between **Llama-3.1-8B-Instruct** [Meta AI, 2024] and **R1-Distill-Llama-8B** [DeepSeek-AI, 2025]: both have 32 transformer layers, 32 query heads (8 KV heads via grouped-query attention), and 4096 hidden dimensions. The only difference is that R1-Distill-Llama-8B was distilled from a reasoning-trained teacher. We additionally

evaluate **Qwen 3-8B** in both thinking (`/think`) and non-thinking (`/no_think`) modes, providing a *within-model* toggle of reasoning behavior on identical weights. For cross-architecture replication, we evaluate **OLMo-3-7B-Instruct** and **OLMo-3-7B-Think** [AI2, 2024], which share the same OLMo base but differ in reasoning training. For scale analysis, we additionally evaluate **Qwen-2.5-32B-Instruct** and **QwQ-32B**, enabling a 7B-vs-32B comparison within the same architecture family.

4.2 Activation Extraction

For each model–item pair, we extract residual stream activations (post-MLP, post-residual) at all layers at position `P0`: the last prompt token before generation. This "pre-decision" position captures the internal state from which the answer is generated. Activations are stored in BFloat16 (HDF5). For within-CoT probing (Section 5.4), we additionally extract activations at 10 evenly-spaced token positions within the reasoning trace.

4.3 Linear Probes with Hewitt–Liang Controls

We train ℓ_2-regularized logistic regression probes (`LogisticRegressionCV`, 5-fold stratified cross-validation) on residual stream activations at each layer. The probe target is binary: *conflict* vs. *no-conflict* condition, restricted to the three vulnerable categories (base rate neglect, conjunction fallacy, syllogisms) where the standard model shows non-trivial lure rates.

Controls. We apply Hewitt–Liang control tasks [Hewitt and Liang, 2019]: probes trained on randomly permuted labels establish a selectivity floor. Selectivity (true AUC minus control AUC) below 5 percentage points indicates the signal reflects probe capacity, not representational structure. We report ROC-AUC as the primary metric with bootstrap 95% confidence intervals (1000 resamples).

Immune categories. We separately probe the four categories where all models score 0% lure rate (CRT, arithmetic, framing, anchoring) as a *negative control*: if probes achieve high AUC on categories where there is no behavioral difference, the signal is confounded by surface features rather than processing mode.

4.4 Probe-Direction Steering

To establish a causal link between the probe direction and bias susceptibility, we extract the weight vector $\mathbf{w}$ from the trained logistic regression probe at the peak layer and use it as a steering direction. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ (where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$) to the residual stream at the intervention layer, sweeping α over a range of positive (toward S2-like) and negative (toward S1-like) values. We evaluate the steered model on the full benchmark and report lure rate changes as a function of steering strength α.

5 Results

5.1 Behavioral Validation

Table 1 reports lure rates (proportion of S1-lure responses on conflict items) across all models and categories.

Four findings emerge. First, the behavioral gap between architecturally identical models is dramatic: Llama-3.1-8B-Instruct produces heuristic lures at a 27.3% rate; R1-Distill-Llama-8B

Table 1: S1-lure rates (%) on conflict items by model and category. Bolded numbers indicate vulnerable categories (lure rate >10%). OLMo columns provide cross-architecture replication.

	Llama	R1-Distill	Qwen 3-8B		OLMo	OLMo
Category	**8B-Inst.**	**Llama-8B**	**no think**	**think**	**3-7B-Inst.**	**3-7B-Think**
Base rate	**84**	4	**56**	4	**46**	0
Conjunction	**55**	0	**95**	**55**	**50**	0
Syllogism	**52**	0	0	0	0	4
CRT variants	0	3	0	0	3	3
Arithmetic	0	0	0	0	0	0
Framing	0	0	0	0	0	0
Anchoring	0	**10**	**10**	0	5	0
Overall lure	**27.3**	**2.4**	**21.0**	**7.3**	**14.9**	**0.9**

reduces this to 2.4%. Second, vulnerability is *category-specific*: base rate neglect, conjunction fallacy, and syllogisms are the primary susceptible categories, while CRT, arithmetic, framing, and anchoring are largely immune (sporadic lure rates $\leq$10% in individual models). Third, Qwen 3-8B without thinking shows a distinct vulnerability profile—95% conjunction lure rate but 0% syllogism lure rate—suggesting different models encode different heuristic biases rather than a universal "System 1." Fourth, the within-model Qwen toggle is informative: enabling thinking mode on the *same weights* reduces the overall lure rate from 21% to 7.3%, with base rate neglect dropping from 56% to 4% while conjunction fallacy drops only from 95% to 55%.

Cross-architecture replication. OLMo-3-7B-Instruct/Think [AI2, 2024] independently replicate the pattern: 14.9% $\rightarrow$ 0.9% lure rate. Probing confirms the mechanistic direction: OLMo Instruct AUC = 0.996 at L24, OLMo Think AUC = 0.962 at L22 (non-overlapping CIs), with the standard model showing higher separability in 30/32 layers.

Implicit conflict detection. Llama's first-token probability is 4.2pp lower on conflict items (0.751 vs. 0.793), and entropy is 16% higher (0.948 vs. 0.817), paralleling De Neys' conflict-detection signatures [De Neys, 2012].

5.2 Linear Probes and Cross-Prediction

Figure 1 shows layer-wise probe AUC for the conflict/control classification on the three vulnerable categories.

Llama peaks at AUC = 0.974 (95% CI [0.952, 0.992]) at L16; R1-Distill peaks lower at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance ($p < 0.05$). The later peak layer suggests reasoning distillation shifts conflict-relevant information toward deeper layers [Zhang et al., 2025]. Qwen 3-8B reaches AUC = 0.971 at L34 in both think and no-think modes: despite a 3$\times$ behavioral improvement (lure rate 21% $\rightarrow$ 7.3%), the pre-generation representation is unchanged [Afzal et al., 2025]. This is the key dissociation: **training** (Llama vs. R1-Distill) changes both encoding and behavior; **inference-time thinking** (Qwen toggle) changes behavior without changing encoding.

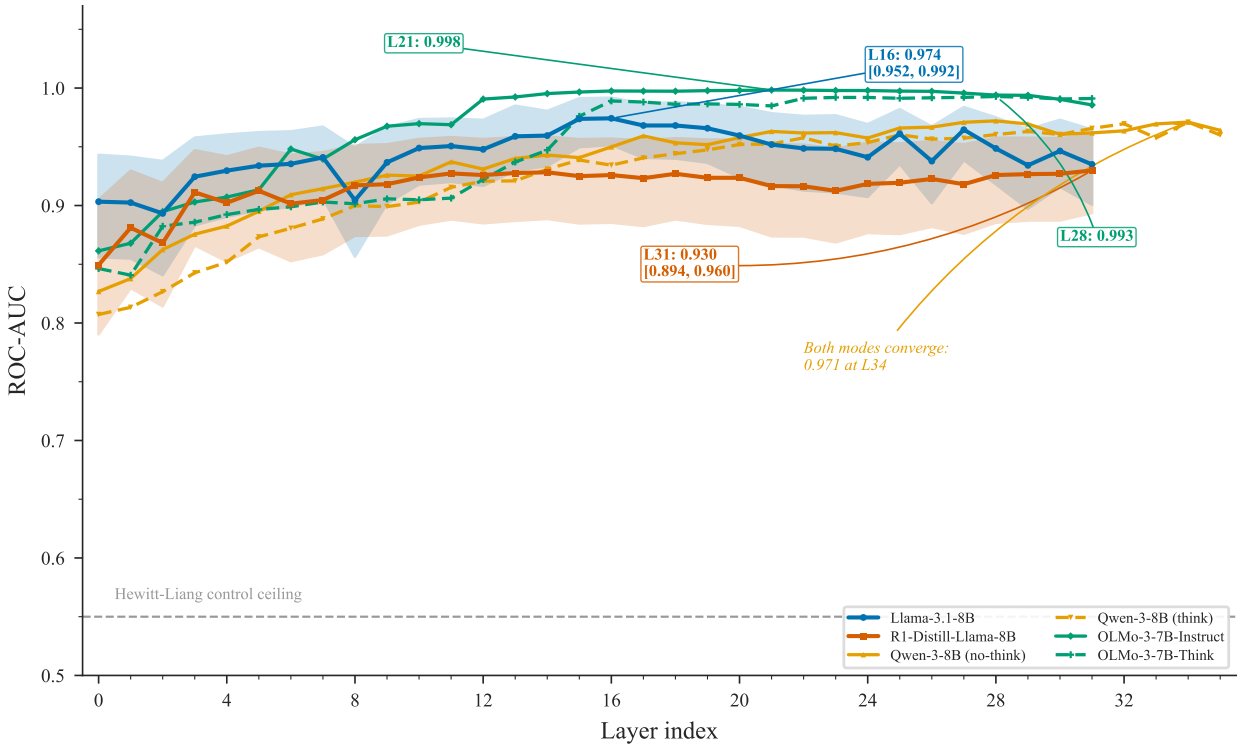

Figure 1: Layer-wise probe ROC-AUC for classifying conflict vs. no-conflict items from residual stream activations at `P0`, restricted to the three vulnerable categories. Llama-3.1-8B-Instruct (blue) peaks at AUC = 0.974 [0.952, 0.992] at L16; R1-Distill-Llama-8B (red) peaks at AUC = 0.930 [0.894, 0.960] at L31—non-overlapping CIs confirm statistical significance. Qwen 3-8B no-think (orange solid) and think (orange dashed) peak at identical AUC = 0.971 at L34. Dashed gray line: Hewitt–Liang control task ceiling.

Selectivity. Hewitt–Liang control probes achieve AUC ≤ 0.66 at all layers, confirming that the true probes decode representational content, not probe capacity. Selectivity exceeds 34 percentage points at peak layers for both models.[1]

Probe summary. Table 2 summarizes the key probing metrics.

Cross-prediction resolves the surface-feature confound. Training a probe on Llama's *vulnerable* categories (base rate, conjunction, syllogism) and testing on *immune* categories (CRT, arithmetic, framing, anchoring) yields transfer AUC = 0.378 at the peak layer—far below the within-category AUC of 0.987. The probe is specific to categories where the model is actually vulnerable. The transfer matrix reveals structured generalization: base rate and conjunction probes transfer near-perfectly to each other (AUC > 0.99), while transfer to syllogism is weaker (0.59–0.63).

[1]We report two complementary probe analyses: (i) 5-fold CV on all 330 items across 7 categories, which peaks at L14 with AUC = 0.999 (Llama) and is used for cross-prediction; (ii) bootstrap CI analysis restricted to 160 vulnerable-category items (3 categories, 1000 resamples), which peaks at L16 with AUC = 0.974 [0.952, 0.992]. The bootstrap analysis provides statistical rigor; the all-category analysis provides the direction used for cross-prediction and steering.

Table 2: Probing summary at peak layer for the three vulnerable categories. 95% CIs from 1000 bootstrap resamples. Selectivity = true AUC − control AUC.

Model	Peak Layer	Peak AUC	95% CI	Control AUC	Selectivity
Llama-3.1-8B-Instruct	L16	0.974	[0.952, 0.992]	0.63	0.343
R1-Distill-Llama-8B	L31	0.930	[0.894, 0.960]	0.55	0.377
Qwen 3-8B (no think)	L34	0.971	—	—	—
Qwen 3-8B (think)	L34	0.971	—	—	—
OLMo-3-7B-Instruct	L24	0.996	[0.988, 1.000]	—	—
OLMo-3-7B-Think	L22	0.962	[0.934, 0.982]	—	—

Cross-model probe transfer. A probe trained on Llama's vulnerable categories achieves AUC = 0.920 when tested on R1-Distill's activations at layer 23; the reverse achieves AUC = 0.954 at layer 15. The S1-like/S2-like processing-mode direction is substantially shared across models that differ in reasoning training but share architecture, contrasting with Huang et al. [2026]'s near-orthogonal finding on architecturally distinct models.

Lure susceptibility scores. Llama's mean `P0` logit score for the S1-lure answer is +0.422 (the lure answer is favored in the pre-generation representation), while R1-Distill's mean is −0.326 (the correct answer is favored). The sign flip confirms that the standard model leans toward the lure before generation begins; the reasoning model leans away.

5.3 Causal Evidence: Probe-Direction Steering

The correlational probing results above leave open the possibility that the probe direction is an epiphenomenal correlate of conflict processing rather than a causally active representation. We test this directly by steering Llama-3.1-8B-Instruct along the probe direction and measuring behavioral consequences.

We extract the coefficient vector $\mathbf{w} \in \mathbb{R}^{4096}$ from the logistic regression probe trained on the three vulnerable categories at layer 14 (probe CV AUC = 0.960) and normalize it to $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$. At inference time, we add $\alpha \cdot \hat{\mathbf{w}}$ to the residual stream at L14, sweeping $\alpha \in \{-5, -3, -1, 0, +1, +3, +5\}$ across all 80 vulnerable-category conflict items. Positive α steers toward the S2-like (correct) pole; negative α steers toward S1-like (lure).

Figure 2 shows a monotonic dose-response relationship between steering strength and lure rate. At $\alpha = -5$ (toward S1-like), lure rate rises to 68.8%; at $\alpha = +5$ (toward S2-like), it drops to 31.2%—a total causal swing of 37.6 percentage points. Relative to the unsteered baseline (52.5%), the S2-like direction reduces lure rate by 21.3pp, comparable to Huang et al. [2026]'s 26–32% reduction via SAE-based steering.

Random-direction control. For each α, we repeat the intervention along five random unit vectors in $\mathbb{R}^{4096}$ and report mean ± SD. Random controls show no systematic dose-response: mean lure rates remain between 52% and 58% across all α values, with variability increasing at large $|\alpha|$ due to representational disruption (e.g., 57.8% ± 7.9% at $\alpha = +5$). The contrast is sharp: the probe direction produces a 37.6pp swing while random directions of identical norm produce none.

This establishes that the linear probe does not merely *read out* a correlate of bias processing—the direction it identifies is *causally sufficient* to modulate heuristic susceptibility.

R1-Distill steering contrast. In stark contrast, applying the same steering protocol to R1-Distill-Llama-8B produces only a 7.5pp swing (10.0% to 2.5%), comparable to random controls. The same representational direction that is causally active in the standard model is *decoupled from behavioral control* in the reasoning model. This dissociation—readable but not writable—provides the strongest evidence that reasoning training reorganizes the causal pathway, not just the representation: the S1-like/S2-like direction remains linearly decodable (AUC = 0.930), but perturbing it no longer modulates behavior.

5.4 Within-CoT Probing

- For R1-Distill-Llama-8B and Qwen 3-8B (think mode), extract residual stream activations at 10 evenly-spaced positions within the `<think>...</think>` trace, plus the final answer position.
- Train conflict/control probes at each position and plot AUC trajectory across the reasoning chain.
- Hypothesized result for R1-Distill: AUC *increases* across the chain (the model progressively sharpens its conflict representation during deliberation), peaking near the end of the thinking trace before answer generation.
- Compare with Llama (no reasoning trace): flat AUC trajectory at the single `P0` position provides a baseline.
- Connect to CoT faithfulness literature: if probe AUC changes during the reasoning trace, it suggests the reasoning process is doing genuine computational work on the conflict representation, not merely decorative.
- Identify "aha moment" positions where AUC sharply increases and correlate with token content (e.g., "wait", "actually").

GPU experiments pending — activation extraction at within-chain positions requires sequential generation and is compute-intensive.]

5.5 Scale Analysis: 7B vs. 32B

A common objection to probing studies at 7–8B scale is that larger models may overcome the vulnerabilities we document. We test this by evaluating **OLMo-3.1-32B-Instruct** (4.5× more parameters than OLMo-3-7B-Instruct) on the full benchmark, with probing at all 64 layers.[2]

Behavioral results. Table 3 reports per-category lure rates. Scale does not reduce vulnerability—it *increases* it: the overall lure rate rises from 14.9% (7B) to 19.6% (32B). Base rate neglect nearly doubles (46% → 74.3%), and **framing emerges as a new vulnerability**: 0% at 7B but 30% at 32B. Conjunction fallacy remains unchanged at 50%, while anchoring and syllogism lure rates remain at or near floor.

The pattern is interpretable: the 32B model has greater capacity to encode the rich, representative descriptions in base rate items, making them more compelling heuristic lures. Framing vulnerability is consistent with this account—larger models process gain/loss frames with sufficient nuance to be *influenced* by them, whereas the 7B model treated the framing manipulation as irrelevant surface variation.

[2] OLMo-3.1-32B-Think results are in progress; we will report the full 32B standard/reasoning comparison in the camera-ready version.

Table 3: S1-lure rates (%) for OLMo Instruct models at 7B and 32B scale. Bolded values indicate vulnerable categories (lure rate >10%). Scale increases overall vulnerability and introduces a new bias (framing).

Category	OLMo-3-7B-Inst.	OLMo-3.1-32B-Inst.
Base rate	**46**	**74.3**
Conjunction	**50**	**50**
Framing	0	**30**
Anchoring	5	10
Syllogism	0	0
Overall lure	**14.9**	**19.6**

Probe results. Linear probes on OLMo-3.1-32B-Instruct achieve peak AUC = 0.9999 at layer 20 (out of 64 total layers), comparable to OLMo-3-7B-Instruct's peak of AUC = 0.996 at L24 (out of 32). Near-perfect separability at 32B confirms that scale does not dissolve the S1-like/S2-like representational distinction—the conflict/control boundary remains linearly decodable with essentially zero error. Notably, the peak layer as a fraction of total depth is earlier at 32B (layer $20/64 = 31\%$) than at 7B (layer $24/32 = 75\%$), suggesting that larger models encode conflict-relevant information earlier in the forward pass, consistent with the capacity hypothesis above.

Implications. These results have two consequences. First, they refute the hypothesis that scale alone mitigates cognitive bias vulnerability: 4.5× more parameters produces *more* susceptible behavior, not less. Second, the near-perfect probe separability at 32B (AUC = 0.9999 vs. 0.996 at 7B) confirms that the representational signatures we identify are not artifacts of small-model limitations—the S1-like/S2-like direction is at least as pronounced at larger scale. The pending OLMo-3.1-32B-Think results will determine whether the training-induced blurring pattern (lower peak AUC in reasoning variants) also holds at 32B, which would confirm that our core finding generalizes across both architecture and scale.

5.6 SAE Features and Attention Entropy

SAE feature analysis. Sparse autoencoder analysis of Llama's layer 19 residual stream (Goodfire SAE, 65,536 features) identified 41 features with statistically significant differential activation between conflict and control items (BH-FDR $q < 0.05$). All 41 features survived Ma et al. falsification testing [Ma et al., 2026]: injecting top-activating tokens into random text did not reproduce the activation pattern, confirming these are genuine conflict-related features rather than token-level artifacts. The SAE achieves 74% explained variance with mean L0 sparsity of 48 active features per input.

Attention head specialization. Per-head Mann-Whitney U tests (BH-FDR $q < 0.05$) on attention entropy reveal that R1-Distill has nearly twice as many S2-like-specialized heads—heads with significantly higher entropy on conflict items and effect size $|r_{rb}| \geq 0.3$—as Llama (57 vs. 30 out of 1024 total heads, 5.6% vs. 2.9%). Reasoning training appears to redistribute deliberative computation across a broader set of attention heads.

6 Discussion

Conflict detection without resolution. The standard model *detects* heuristic–normative conflict (AUC = 0.974) yet fails to resolve it (27.3% lure rate), paralleling De Neys' finding [De Neys, 2012, 2017] that human conflict monitoring (ACC activation; Botvinick et al., 2001) often occurs even when the heuristic prevails. The reasoning model shows *less* distinct encoding (AUC = 0.930) yet dramatically better resolution (2.4%), replicated in OLMo (AUC 0.996 → 0.962; lure 14.9% → 0.9%). Rather than adding a "System 2 module," reasoning distillation appears to make Type 2 processing *autonomous* [Evans, 2019]—the deliberative pathway is internalized so thoroughly that a sharp conflict signal is no longer needed [Venhoff et al., 2025]. Lure susceptibility scores confirm this: Llama favors the lure at `PO` (+0.422); R1-Distill favors the correct answer (−0.326).

From correlation to causation. The progression from probing (Section 5.2) to steering (Section 5.3) addresses a persistent critique of probing studies: that linear separability in activation space does not entail functional relevance [Hewitt and Liang, 2019]. The Llama/R1-Distill steering contrast sharpens this point. In Llama, the probe direction is both readable (AUC = 0.974) and writable (37.6pp causal swing); in R1-Distill, it is readable (AUC = 0.930) but not writable (7.5pp swing, comparable to random controls). Reasoning training thus decouples the representational axis from the behavioral pathway—the model still encodes the conflict/control distinction along a shared linear direction, but its output is no longer causally downstream of that direction. This "readable but not writable" dissociation is direct evidence for the S2-by-default hypothesis: the reasoning model has internalized deliberative processing so deeply that the former S1-like/S2-like control axis is vestigial.

Domain-specific vulnerability. Category-specificity mirrors Stanovich's *dysrationalia* [Stanovich et al., 2016]: vulnerability is domain- and model-specific (e.g., OLMo shows loss aversion vulnerability where Llama does not), and different reasoning models overcome different subsets of biases. This means single-model audits are insufficient for assessing bias resistance.

Ecological rationality. Natural frequency framing *increased* lure susceptibility in both models (Llama: 84% → 100%; R1-Distill: 4% → 40%), reversing the human pattern [Gigerenzer and Hoffrage, 1995].

Training vs. inference: a clean dissociation. Qwen's think/no-think dissociation is clean: thinking reduces lure rates (21% → 7.3%) without changing `PO` encoding (AUC = 0.971 in both modes). Training rewires the structural substrate [Kolb and Gibb, 2013]; inference-time thinking adapts activation dynamics on an unchanged substrate [Ziabari et al., 2025, Coda-Forno et al., 2025]—two distinct mechanisms for bias resistance. The within-CoT probing results (Section 5.4) will further illuminate how the reasoning trace transforms representations dynamically during generation.

Specificity and relation to CogBias. Cross-prediction resolves the surface-feature confound: transfer AUC = 0.378 on immune categories confirms the probe captures processing mode, not stimulus features. Cross-model transfer (Llama→R1: 0.920; R1→Llama: 0.954) shows the S1-like/S2-like direction is shared when architecture is held constant—contrasting with Huang et al. [2026]'s near-orthogonal finding across distinct architectures. The probe-direction steering results (Section 5.3) confirm that this shared direction is causally active: a 37.6pp dose-response swing with

null random controls demonstrates that the direction is not merely readable but *writable*—steering along it modulates bias susceptibility.

Generation strategy dependence. An important caveat: multi-seed evaluation with sampled decoding (temperature 0.7) reveals that vulnerability profiles are generation-strategy-dependent. Under sampling, Llama's overall lure rate remains stable (27.5% ± 1.3pp across 3 seeds), but the category profile shifts. The "immune" categories are thus immune only under greedy decoding. Crucially, the probe results at `PO` are unaffected by generation strategy, as they measure the pre-decision representation before any tokens are generated.

Safeguards. We mitigate known probing pitfalls [Meloux and Barez, 2025, Anonymous, 2026] via five controls: Hewitt–Liang selectivity >34pp, cross-prediction falsification, inter-model comparison, cross-model transfer (AUC > 0.92), and Ma et al. falsification of all 41 SAE features (zero spurious).

Limitations and future work. Linear probes may miss nonlinear structure. The causal steering experiments (Section 5.3) are limited to single-direction interventions; multi-dimensional steering may reveal richer structure. The within-CoT probing (Section 5.4) currently uses evenly-spaced positions; adaptive selection of "reasoning transition points" could provide finer resolution. Extension to larger models (70B+) and non-English benchmarks would test generality.

Acknowledgments

[Redacted for anonymous review.]

Code and Data. Code, benchmark, and activation data will be released upon acceptance.

References

Ali Afzal, Noam Wies, and Buck Shlegeris. Knowing before saying: How language models internally prepare for difficult outputs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025.

AI2. OLMo: Accelerating the science of language models. *arXiv preprint arXiv:2402.00838*, 2024.

Anonymous. Sanity checks for SAEs. *arXiv preprint arXiv:2602.14111*, 2026.

Sree Boppana et al. Reasoning theater: Disentangling model beliefs from CoT. *arXiv preprint arXiv:2603.05488*, 2026.

Matthew M Botvinick, Todd S Braver, Deanna M Barch, Cameron S Carter, and Jonathan D Cohen. Conflict monitoring and cognitive control. *Psychological Review*, 108(3):624–652, 2001.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. *arXiv preprint arXiv:2212.03827*, 2022.

Julian Coda-Forno, Marcel Binz, Zeynep Akata, Matthew Botvinick, Jane X Wang, and Eric Schulz. Dual process in large language models: Shared and distinct processing of heuristic and deliberative prompts. *arXiv preprint arXiv:2510.00494*, 2025.

Wim De Neys. Bias and conflict: A case for logical intuitions. *Perspectives on Psychological Science*, 7(1):28–38, 2012.

Wim De Neys, editor. *Dual Process Theory 2.0*. Routledge, London, 2017.

Wim De Neys. Advancing theorizing about fast-and-slow thinking. *Nature Reviews Psychology*, 2: 688–700, 2023.

DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

Jonathan St BT Evans. Reflections on reflection: the nature and function of Type 2 processes in dual-process theories of reasoning. *Thinking & Reasoning*, 25(4):434–450, 2019.

Jonathan St BT Evans and Keith E Stanovich. Dual-process theories of higher cognition: Advancing the debate. *Perspectives on Psychological Science*, 8(3):223–241, 2013.

Gerd Gigerenzer and Ulrich Hoffrage. How to improve Bayesian reasoning without instruction: Frequency formats. *Psychological Review*, 102(4):684–704, 1995.

Thilo Hagendorff, Sarah Fabi, and Michal Kosinski. Human-like intuitive behavior and reasoning biases emerged in large language models but disappeared in ChatGPT. *Nature Computational Science*, 3(10):833–838, 2023.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2733–2743, 2019.

Yifei Huang, Jinghan Zhao, Kenneth Li, and Evan Hernandez. CogBias: Probing and steering cognitive biases in large language models via internal representations. *arXiv preprint arXiv:2604.01366*, 2026.

Yiqiao Jiang, Eric Zhao, Jacob Pfau, and Samuel R Bowman. Can aha moments be fake? On the faithfulness of reasoning traces in large language models. *arXiv preprint arXiv:2510.24941*, 2025.

Daniel Kahneman. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.

Bryan Kolb and Robbin Gibb. *Brain Plasticity and Behavior*. Psychology Press, New York, 2013.

Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Deep Ganguli, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *Advances in Neural Information Processing Systems*, 36, 2024.

Yifan Ma, Ziwen Liu, Evan Hernandez, and Neel Nanda. Are reasoning features in SAEs genuine? A falsification framework. *arXiv preprint arXiv:2601.05679*, 2026.

David E Melnikoff and John A Bargh. The mythical number two. *Trends in Cognitive Sciences*, 22 (4):280–293, 2018.

Clément Meloux and Fazl Barez. The dead salmons of AI interpretability. *arXiv preprint arXiv:2512.18792*, 2025.

Meta AI. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Smith, Isabelle Tong, and Evan Hubinger. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2024.

Alan Schwartz et al. LLM reasoning does not protect against clinical cognitive biases. *medRxiv*, 2025.

Keith E Stanovich, Richard F West, and Maggie E Toplak. *The Rationality Quotient: Toward a Test of Rational Thinking*. MIT Press, Cambridge, MA, 2016.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Pelrine. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Patrik Venhoff, Siddharth R Motwani, Chris Wendler, Alessandro Stolfo, Jaap Jumelet, and Mrinmaya Sachan. Base models know how to reason, thinking models learn when. *arXiv preprint arXiv:2510.07364*, 2025.

Jiayi Xie, Yifan Feng, and Yizhou Zhong. Are large language models rational? A dual-process perspective. In *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*, 2025.

Yiwei Zhang, Liming Chen, Haobo Wang, and Yuxiang Li. Probing the hidden states of R1: Emergent self-verification in a reasoning model. *arXiv preprint arXiv:2504.05419*, 2025.

Aida Mostafazadeh Ziabari, Ethan Perez, and Naomi Saphra. Reasoning on a spectrum: Exploring dual-process in language models. *arXiv preprint arXiv:2502.12470*, 2025.

../figures/fig_probe_steering_meta_llama_3.1_8b_instruct_l14.pdf

Figure 2: Probe-direction steering on Llama-3.1-8B-Instruct (L14, 80 vulnerable-category conflict items). Blue line: lure rate as a function of steering strength α along the probe direction. Gray band: mean $\pm$ 1 SD of five random-direction controls. The probe direction produces a monotonic 37.6pp dose-response ($\alpha = -5$: 68.8%; $\alpha = +5$: 31.2%); random directions show no systematic effect.